·Solutions·

1.

1. When all rays just reach the sphere, the edge ray is tangent at T (Fig. 1); geometry gives

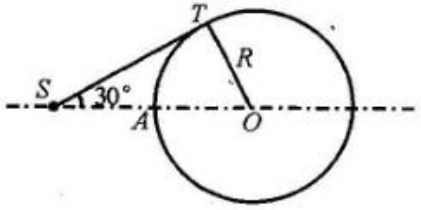


Fig. 1

$$\sin 30^\circ = \frac{R}{\overline{SO}}$$

Hence

$$\overline{SA} = R$$

If the source is over R from A, some rays miss.

The whole beam reaches the cap if source - A distance obeys

$$x \leqslant R \tag{1}$$

2. For $x = R$, a ray at α to the axis: Fig. 2.

B is on the axis; dashed QBQ' and QAQ' form a sphere. θ and θ' are incidence/refraction angles; intersection: P'.

β: OP' - axis angle; α': refracted ray - axis angle.

The sine rule for $\triangle OPS$ gives

$$\frac{2R}{\sin\theta} = \frac{R}{\sin\alpha} \tag{2}$$

The refraction law gives

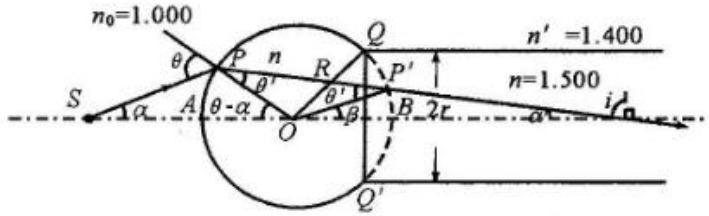


Fig. 2

$$\sin\theta = 1.5\sin\theta' \tag{3}$$

Geometry gives

$$\beta = 2\theta' - \theta + \alpha \tag{4}$$

$$\alpha' = \theta - \alpha - \theta' \tag{5}$$

If its backward extension meets the axis: Fig. 3. Set the sign of α' as follows: a ray bent towards the axis gives

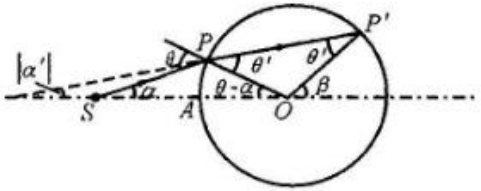


Fig. 3

positive α'. If it bends away (the backward extension meets the axis left of A), α' is negative; equation (5) still holds.

Geometry also gives the fibre-entry condition:

$$|\beta| \leqslant \angle QOB \tag{6}$$

β may be negative, meaning P' is below the axis.

and $$\angle QOB = \arcsin\frac{r}{R}$$